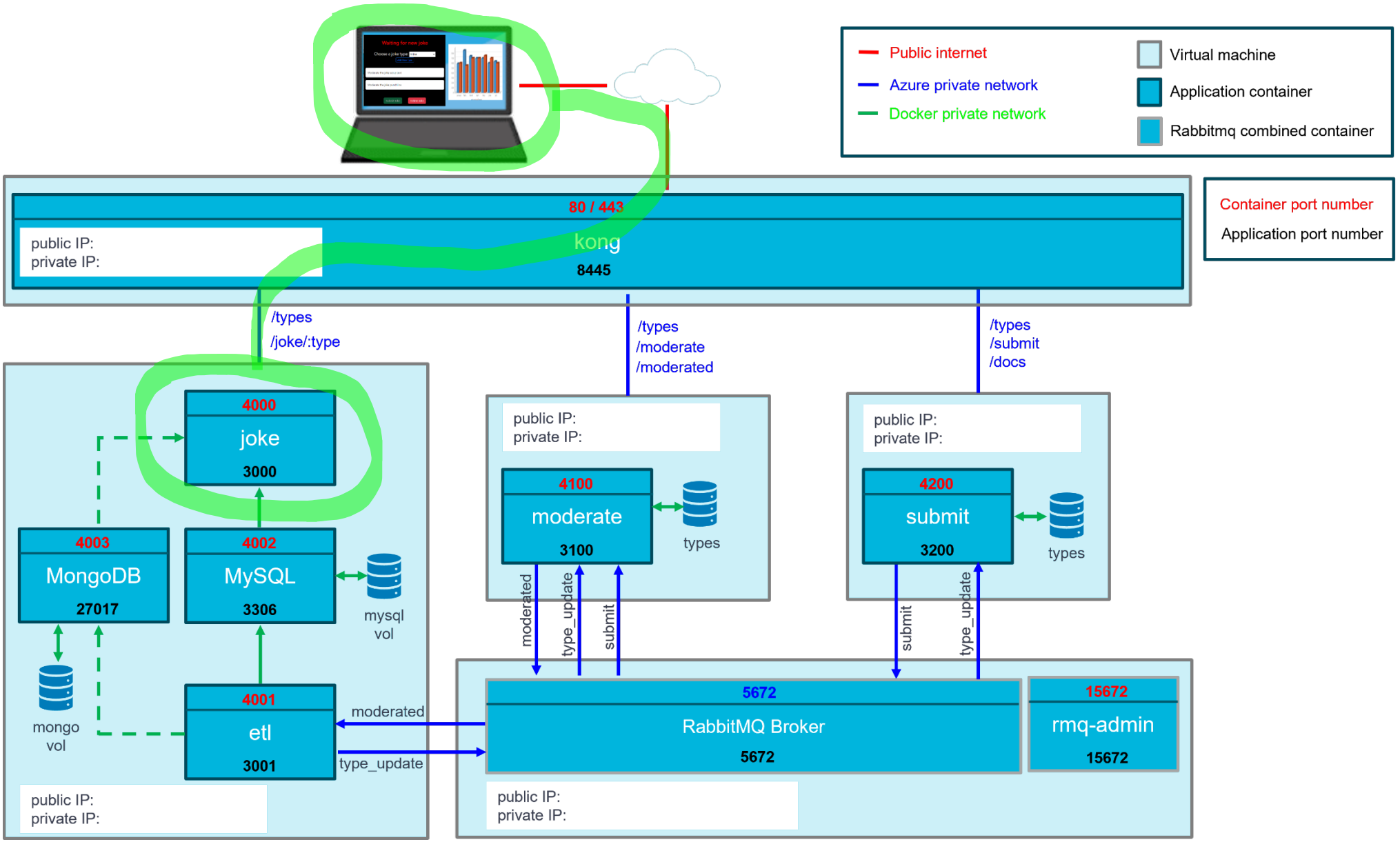
Lab 3 Static Web Server, App Server and Front-end integration



Practical Exercise: Integrate frontend with the backend

Based on the demos, particularly **fetchDemo**, and previous lab exercises, create a web API to serve up jokes using static web server middleware and the frontend building on your previous lab solution of a joke API, linking the front-end to the backend to create a single integrated solution where the web assets and data are provided from the backend and rendered in the browser.

Additional requirement:

Add a dropdown to your frontend

Implement JavaScript to call your backend **GET /types** endpoint

The returned types should be used to populate the dropdown

When the user selects a joke type and presses the Get Joke button, display the setup and punchline if one joke is requested or output a list if more than one is selected. i.e., the same functionality as before but the data source is no longer a fixed array in the frontend JavaScript but one in at the backend

When displaying a single joke, add a 3 second delay before the punchline is displayed using the setTimeout function.

The following is an example. Don't spend ages making it look nice, it's the concepts that are important

You can check how random your jokes are when looking at the IDs in the displayed table.

As usual, if anything is unclear or you get stuck then contact me on teams.


Joke Central
Choose a joke type: Dad
Number of jokes: 1
Get Joke
Why do fathers take an extra pair of socks when they go golfing?
In case they get a hole in one!


Joke Central
Choose a joke type: Knock Knock
Number of jokes: 100
Get Joke
ID Type Setup ID
12 knock-knock Knock knock. Who's there? A broken pencil. A broken pencil who? Never mind. It's pointless.
13 knock-knock Knock knock. Who's there? Cows go. Cows go who? No, cows go moo.
34 knock-knock Knock knock. Who's there? Opportunity. That is impossible. Opportunity doesn't come knocking twice!
14 knock-knock Knock knock. Who's there? Little old lady. Little old lady who? I didn't know you could yodel!
61 knock-knock Knock knock. Who's there? Hatch. Hatch who? Bless you!